

Load Interconnection System Impact Study Report

For Queue Project Q13-0284

East 75Th Street 138 kV

April 20, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0284. The Interconnection Customer proposes a 39.9 MW large load (data center) facility interconnecting at the East 75Th Street 138 kV substation in the Pennsylvania area. The request was received on November 10, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 182 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$77,294,000.

3. Network Impacts

3.1 Load Deliverability Analysis

S HAMPTON – RAILYARD 138 KV: 139.0% loading (287 MVA) following the outage of the WALLENPAUPACK 500/345 kV Auto; DFAX = 9.9%.

The WEST GALLATIN – MEREDOSIA 138 KV line is overloaded to 131.2% of its 261 MVA emergency rating for the loss of the W GALLATIN – FELIDA 345 KV. The project contributes 8.5% DFAX (3.4 MW) to this overload.

The LAKIN MUNICIPAL – S HAMPTON 138 KV is overloaded to 128.4% of its 316 MVA emergency rating for the loss of the WALLENPAUPACK 500/345kV XFMR. The project contributes 9.9% DFAX (4.0 MW) to this overload.

The CENTRAL CAMPUS – FIBERBOARD 138 KV is overloaded to 125.4% of its 197 MVA emergency rating for the loss of the ALDERTON – CLEAR SPRINGS 345 KV L. The project contributes 21.0% DFAX (8.4 MW) to this overload.

The FELIDA 500/345 kV Auto is overloaded to 117.4% of its 873 MVA emergency rating for the loss of the TALENENERGY MARTINS CREEK LLC HARWOOD – FELIDA 345 KV line. The project contributes 12.1% DFAX (4.8 MW) to this overload.

The RAILYARD – CENTRAL CAMPUS 138 KV is overloaded to 117.0% of its 223 MVA emergency rating for the loss of the ALDERTON – CLEAR SPRINGS 345 KV. The project contributes 21.0% DFAX (8.4 MW) to this overload.

For the loss of EAST 75TH STREET SUBSTATION – BAR BEACH 138 KV, the FIBERBOARD – WARRICK 138 KV line loads to 105.7% of its 209 MVA rating. Project DFAX: 5.8%.

WALLENPAUPACK – HILLS CHAPEL 345 KV: 102.9% loading (1,212 MVA) following the outage of the WALNUT – W GALLATIN 345 KV line; DFAX = 6.3%.

The CONNELL – TOMBSTONE 345 KV is overloaded to 101.5% of its 1,138 MVA emergency rating for the loss of the TOMBSTONE – MARCUS HOOK ENERGY LP SUBSTATION 500 KV. The project contributes 5.5% DFAX (2.2 MW) to this overload.

For the loss of TOMBSTONE – MARCU HOOK ENERGY LP 500 KV, the TOMBSTONE 500/345kV Transformer loads to 100.2% of its 940 MVA rating. Project DFAX: 5.3%.

CONNEL – CHIQUITA 345 KV: 100.0% loading (1,040 MVA) following the outage of the TOMBSTONE – MARCUS HOOK ENERGY LP 500 KV L; DFAX = 5.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
S HAMPTON – RAILYARD 138 KV line	Rebuild line with higher-capacity conductor	\$5,995,000	100% (\$5,995,000)
WEST GALLATIN – MEREDOSIA 138 KV line	Rebuild line with higher-capacity conductor	\$3,022,000	15% (\$464,000)
S HAMPTON – LAKIN MUNICIPAL 138 KV L	Rebuild line with higher-capacity conductor	\$18,337,000	100% (\$18,337,000)
CENTRAL CAMPUS – FIBERBOARD 138 KV	Rebuild line with higher-capacity conductor	\$41,445,000	100% (\$41,445,000)
FELIDA 500/345 kV Auto	Replace transformer with higher-rated unit	\$22,980,000	3% (\$728,000)
RAILYARD – CENTRAL CAMPUS 138 KV	Rebuild line with higher-capacity conductor	\$8,255,000	100% (\$8,255,000)
FIBERBOARD – WARRICK 138 KV	Reconductor line	\$1,738,000	100% (\$1,738,000)
WALLENPAUPACK – HILLS CHAPEL 345 KV	Reconductor line	\$16,369,000	1% (\$129,000)
CONNELL SUBSTATION – TOMBSTONE SUBSTATION 345 KV	Reconductor line	\$9,604,000	0% (\$40,000)
TOMBSTONE SUBSTATION 500-345 kV XFMR	Replace transformer with higher-rated unit	\$22,508,000	0% (\$109,000)
CONNEL – CHIQUITA 345 KV	Reconductor line	\$13,186,000	0% (\$54,000)
	Total allocated to project		\$77,294,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.